

COMS20017 – Algorithms & Data

Problem Sheet MM03

1 – Using $\sin(2\pi nx)$, demonstrate the concept of superposition as follows:

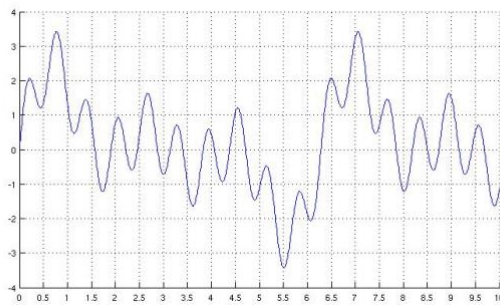
- first plot three sine functions over the range ± 3 in steps of 0.1 using $n=\{1/4, 1, 2\}$. Note, plots should appear in the same graph to give a better sense of what is happening.
- Now plot in a different colour the sum of all the sines above.
- Add more sine functions over the same range and repeat step (b).

Matlab:

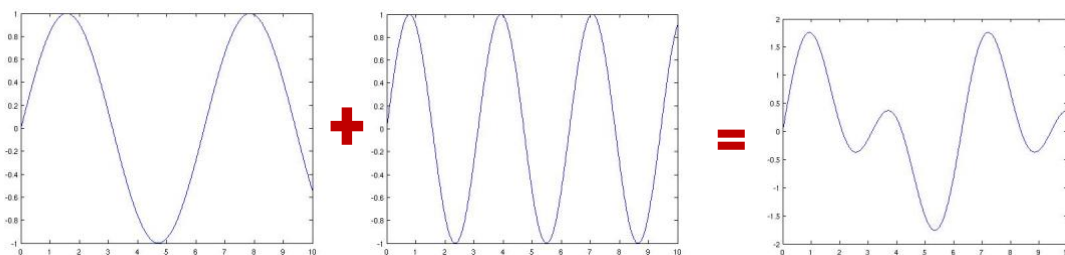
- First define the range, say $x = [-3:0.1:3]$
The sine function plot over the specified range with $n=1/4$ is then `plot(sin(2*pi*x*1/4))`
Hold the plot. Now plot again for the other values of n .
- Add the sines from (a) and plot the new function using 'r' as a parameter of the plot function to draw in red. See *help plot* if unsure of the syntax.

Python: see `sines.py` on unit github page

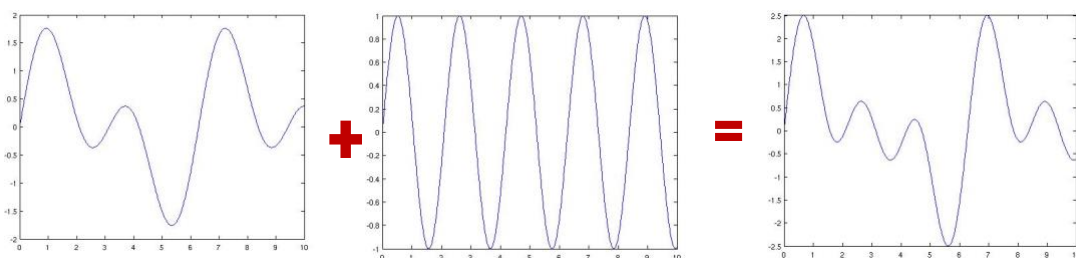
2 – Based on your understanding of the Nyquist Sampling Rate theorem, what is a sufficient sampling rate for the signal below? Hint: the signal is composed of the summation of $\sin(x)$, $\sin(2x)$, $\sin(3x)$ and $\sin(10x)$.



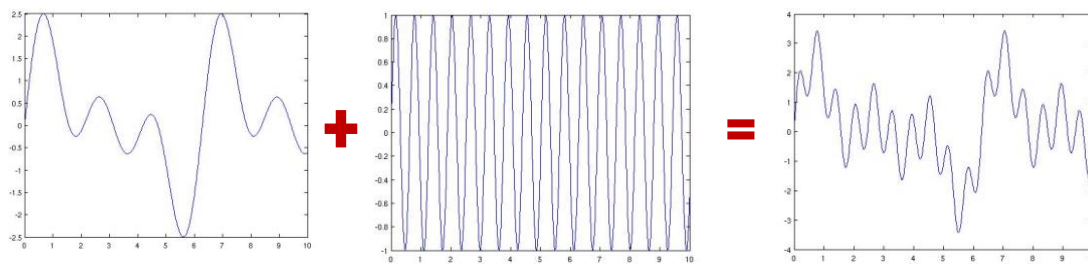
Consider the different frequency components that were used to build the signal (like in Fourier Analysis). This shows the waves $\sin(x)$ and $\sin(2x)$ and their sum:



When adding the summed wave $\sin(x)+\sin(2x)$ to the higher frequency wave $\sin(3x)$ then the wave to the right results:



Next we add $\sin(x)+\sin(2x)+\sin(3x)$ to $\sin(10x)$, resulting in the signal.



The highest frequency in the figure is thus that of the wave $\sin(10x)$. The frequency is thus $10/(2\pi) = 1.59$ Hz. Following the Nyquist theorem, the sampling rate should be at least 3.18 Hz (2×1.59).

3 – Determine which is an even and which is an odd function:

- (i) $f(x) = 7x^3 - x$ odd
- (ii) $f(x) = 3x^2 + 1$ even
- (iii) $f(x) = 3x^2 \sin(x)$ odd
- (iv) $f(x) = \frac{3}{(-x)^4 - 4}$ even
- (v) $f(x) = \cos(x) + 5x - 3$ No symmetry of any kind, so it is neither even nor odd.

4 – The period of the signal $x(t) = 10 \sin 12\pi t + 4 \cos 18\pi t$ is:

- a) $\pi/4$
- b) $1/6$
- c) $1/9$
- d) $1/3$
- e) $1/30$

Factor out 2π . Then, there are two waveforms of frequencies 6 and 9, respectively. Hence, the combined frequency is the highest common factor between 6 and 9 which is 3. The period is then $1/3$.

5 – The following gene sequence contains significant frequencies. Design two different symbolic encodings and in each case apply your encoding to extract some of these frequencies.

ACAGAGATACAGAGATACAG

$A=1, G=C=T=0 \rightarrow 10101010101010101 \dots$ so period is 2, $f=1/2$
 $A=1, G=2, C=3, T=4 \rightarrow 12131314121313141213 \dots$ so period is 8, $f=1/8$

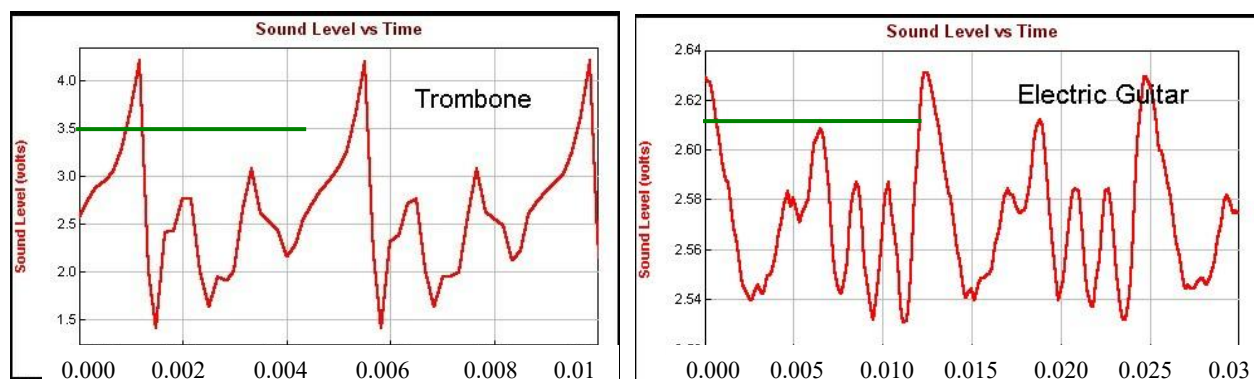
6 – If the fastest oscillations that we want to measure are at 120 Hz, which of the following is the most reasonable sampling rate?

- a. 60 Hz
- b. 60 kHz
- c. anything over 0.00833 Hz
- d. 250 Hz
- e. 120 Hz

Answer is d. We must sample at, or more than, twice the fastest oscillation in the measured signal.

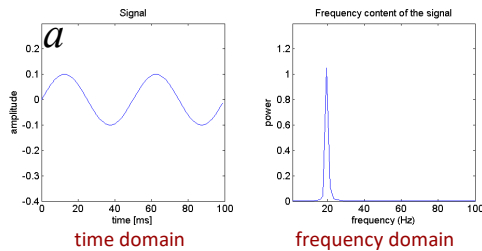
7 – The graphs below display the amplitude of the sound wave for a Trombone and an Electric Guitar as a function of time. The y-axis is the amplitude axis and the x-axis is the time axis. Notice that each one is plotted over a different length of time.

- (a) Mark the period of the signal for each instrument.
- (b) Approximately, how many periods are shown in these graphs for each instrument?
- (c) Approximately, what is the peak amplitude in each case?
- (d) Approximately, what is the frequency given the signal period in each case?
- (e) Which signal contains higher frequency information? Why?

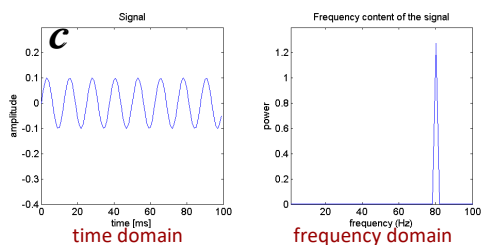
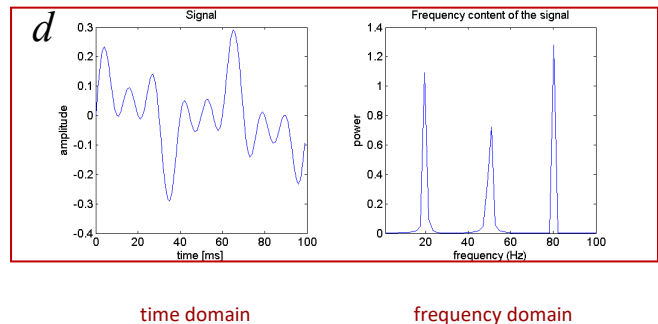
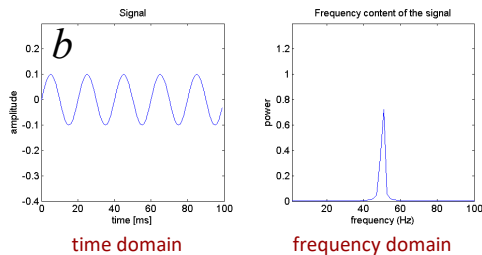


- (a) Marked in Green in the diagram above, about 0.0045 and 0.012 respectively.
- (b) In both cases around 2 and a bit.
- (c) Trombone: about 4.2 EG: about 2.63
- (d) $f = 1/T$ so $1/0.0045 = 222.2$ and $1/0.012 = 83.3$ respectively.
- (e) The Trombone as it cycles more frequently than the EG over the same time period.

8 – Consider the three signals a , b , and c below, and their addition d .



$$d = a + b + c$$



- What would the frequency of the signal d look like?
- How many oscillations per second does signal a have?
- How can you determine the frequency of signal c if you did not have the frequency domain plot of that signal?

- The frequency of signal d would simply include the frequencies of the constituent sinusoids.
- Signal a has a peak frequency of 20Hz, so there are 20 oscillations per second.
- Looking at the time domain plot of the signal, we can count that it repeats around 8 times per 100ms, so it repeats 80 times in 1 second, and so it's an 80Hz signal.

9 – What are the two 1D filters that can replace the 2D filter for W , if they were applied consecutively? Do the same for X .

$$W = \frac{1}{9} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} x \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} x \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}$$

10 – In the Fourier Series of function $f(x) = x$, $-2 < x < 2$, the Fourier coefficient a_2 is equal to:

- 0
- 2
- 1
- 2

It appears that $f(x)$ is odd, since $f(-x) = -x = -f(x)$. Given the function is odd, therefore all a_n terms are 0.